



Badger *Meles meles* setts and bryophyte diversity: A newly found role for the game animal in European temperate forests



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ABSTRACT

Badgers (*Meles meles*) inhabiting Central European forests may be a leading factor driving changes in plant species community and diversity. The digging activity of badgers may shape plant species communities in important ways, especially in managed forests where natural events on the forest floor (e.g. soil disturbance due to tree uprooting) are largely eliminated by silviculture. We tested the following hypotheses in our study: that badger burrows act as focal sites which allow many bryophyte species to occur and complete their reproductive cycle; that the effect of burrows as focal sites of bryophyte diversity is more pronounced in mesotrophic forests than in fertile forests; and that the bryophyte communities of badger burrows differ quantitatively and qualitatively from those of their near vicinity. The study was done in 2013 in the Kampinos Forest, one of the largest (ca. 385 km²) lowland forest complexes in central Poland. We distinguished two types of research sites: badger burrows (main setts) and paired reference plots. The pairs of plots were divided into two groups depending on habitat fertility (C/N ratio): lower-fertility plots ($N = 20$) and higher-fertility plots ($N = 20$). All data were collected from circular plots (5.64 m radius); the burrow plots were centered on the badger burrows and were paired with same-size reference plots. We recorded 55 bryophyte species in the survey, 26 of them exclusively on burrows. The beta diversity of the bryophyte assemblages, expressed as total dissimilarity, turnover and nestedness, was similar for the burrows and the reference plots. The mean number of bryophyte species and their Shannon H' diversity index were significantly higher for burrow plots than for undisturbed reference plots. Neither the number of species nor the species diversity of bryophytes were affected by the interaction of habitat fertility. We observed no relation between the profile of bryophyte groups on the burrows and specific substrate, acidity or life history. Only epiphytic bryophytes were overrepresented in burrow plots. The spatial scale and range of disturbance caused by badger digging suggest that badger setts contribute to the diversity of bryophyte species in managed and protected forests in Central Europe. The burrowing activity of badgers seems to be an important factor boosting the diversity of bryophyte species in managed forests, especially where natural topsoil disturbances are limited by silviculture practices. In forest ecosystems, badger activity creates a patch distribution of microhabitats for vascular plants – and also for bryophytes, as shown here.

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1. Introduction

In temperate forests, small, patchily distributed disturbances (gaps, uprooted trees, pedturbations) are thought to be the main factors influencing soil and vegetation properties, driving changes in species composition (Runkle, 1982; Scharenbroch and Bockheim, 2007) by modifying light, heat and moisture conditions (Borchard et al., 2008; Canals and Sebastiá, 2000; Fedriani et al., 2015; Goszczyńska and Goszczyński, 1977; Hobbs and Mooney,

1985; Kerley et al., 2004; Wilson and Tilman, 1993) as well as nutrient availability (Kurek et al., 2014a) and interspecific competition (Kinlaw, 1999; Wilson and Tilman, 1993). As a result of forest management practices adapted to the timber market, European temperate forests (mostly coniferous monocultures) are now largely devoid of disturbances such as gaps or uprooted trees. Natural disturbances driving species diversity are mostly limited to the primeval forest landscape; they very rarely occur in managed forests (Faliński, 1978). As small-scale disturbances in managed forests are under strong control, burrowing mammals may be one of the few remaining disturbance factors driving changes in species community and diversity on the forest floor. There is every reason to

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believe that burrowing animals play a major part in shaping the species diversity of temperate forests in Europe.

Among burrowing animals, the Eurasian badger (*Meles meles*) is one of the most widespread carnivores on the continent, known for digging extensive, elaborate burrows, or setts, which vary considerably in internal structure and in use. Permanently occupied setts are defined as main setts; they can have many entrances, tunnels and chambers, enlarged from year to year and inherited by many generations over decades (Kowalczyk et al., 2004; Kruuk, 1978; Neal and Roper, 1991). Sett digging produces large amounts of excavated soil, up to 28 m³ in volume and covering up to 740 m² or even more (Coombes and Viles, 2015; Neal and Roper, 1991). As badgers live in social groups and inhabit the same burrows as places for resting, breeding and winter sleep over the course of several years, highly intensive soil exploitation and disturbance are key features of main setts, which have sandy heaps at the entrance. Setts add to habitat heterogeneity in forests and, in consequence, they increase plant species diversity (Kurek et al., 2014a).

Local small-scale disturbances caused by fossorial mammals can substantially influence ecosystem structure and functioning. Pedturbations support plant communities that differ qualitatively and quantitatively from the matrix landscape in which the patches are imbedded (Kurek et al., 2014a). Soil disturbance by fossorial animals elicits a variety of responses by herbaceous vegetation, reflected in species richness, biomass, plant abundance and plant density (Alkon, 1999; Whitford and Kay, 1999). At a wide range of latitudes, digging disturbance usually is noted as having a positive influence on the richness of herbaceous species (Bruun et al., 2005; Kurek et al., 2014a; Obidziński and Głogowski, 2005; Obidziński and Kiełtyk, 2006), an effect which includes the establishment and spread of exotic plant species (Kiełtyk and Mirek, 2015; Kotanen, 1995; Whitford and Kay, 1999).

The majority of research on the effects of burrowing by animals has been focussed on arid and semi-arid environments and/or the impact of burrowing on soil properties and herbaceous plant communities (Eldridge and Myers, 2001; Eldridge and Whitford, 2009; Garkaklis et al., 2004; Kerley et al., 2004; Whitford and Steinberger, 2010). There is an evident lack of similar studies for groups of organisms other than herbaceous plants, such as bryophytes, in temperate forests. The research done for other biomes and plant groups led us to infer that burrows must also be important for bryophytes, particularly since some studies show that the ground-layer cover changes between burrows and their surroundings (Kurek et al., 2014a; Obidziński and Głogowski, 2005).

This paper reports an effect of badger burrows on the species richness and structure of bryophyte communities, in a study comparing burrows with reference areas. We tested the following hypotheses: that badger burrows act as focal sites which allow many bryophyte species to occur and complete their reproductive cycle; that the effect of burrows as focal sites of bryophyte diversity is more pronounced in mesotrophic than in fertile forests; and that the bryophyte communities of badger burrows differ quantitatively (species abundance) and qualitatively (characteristic species associated only with burrow plots) from surrounding areas.

2. Methods

The study was done in the Kampinos Forest, one of the largest (ca. 385 km²) and best-preserved lowland forest complexes in central Poland. It is situated in the Vistula River Valley, north and west of Warsaw (52.26–52.40°N, 20.28–20.88°E; 68–106 m a.s.l.). As in other parts of Poland, a transitional climate (with continental and marine influences) dominates here, with weather conditions being highly variable across months and years. The growing season

lasts 185 days, average annual temperature is 7.7 °C, and average annual precipitation amounts to 546 mm (Andrzejewski, 2003). Sand of glaciofluvial and fluvial origin is the dominant substrate. A large part of the terrain is formed by wind-blown sand, creating dunes overgrown mostly with pine and mixed oak-pine forest. Moist habitats such as peat bogs and marshy forests with *Alnus glutinosa* occur between the dunes.

We surveyed badger burrows over the entire forest area during the 2009–2011 spring seasons. The burrows were identified as badger setts from observations of tracks, evidence of burrow cleaning, paths, and the animals themselves (Kowalczyk et al., 2000; Panek and Bresiński, 2002). Only breeding sites (main setts) were analyzed in this study. In 2013 we selected the 20 best-developed burrows noted to be in use as breeding sites. The protocol used two types of research sites: burrows and paired reference plots ($N = 40$). Reference plots representing the same type of habitat but having no visible signs of animal-caused disturbance were established 50 m away in a random direction. All data were collected from circular plots (5.64 m radius); the burrow plots were centered on the badger burrows and were paired with same-size reference plots. Bryophyte samples were identified in the laboratory. The abundance of bryophyte species in the ground layer was estimated as percentage of cover for each plot. The total cover of bryophytes in the ground layer (applied in indicator species analysis and in DCA) was estimated visually to the nearest 5%. If the cover of a given species was less than 5%, it was estimated to the nearest 1%. To detect the effect of the burrow environment on the community of bryophyte species, we identified some eco-functional groups after Dierssen (2001) with some modifications, including acidity (h acidophyte – highly acidic – pH 3.4–4.0; c acidophyte – considerably acidic – pH 4.1–4.8; m acidophyte – moderately acidic – pH 4.9–5.6; subneutrophyte – pH 5.7–7.0; basiphyte – pH > 7.0), substrate type (epigeic – growing on soil; epiphytic – on bark of living trees; lignicolous – on decorticated wood of dead trees in different stages of decay; sandy soils – adapted to burial; humicolous – soils rich in humus) and life history (annuals – potential life span of less than one year; short-lived shuttle species and colonists – potential life span of a few years; perennials – potential life span of many years). Bryophyte nomenclature follows Ochrya et al. (2003) and Szwejkowski (2006). For soil physicochemical properties we used data collected in 2012 from the same undisturbed reference plots ($N = 20$) to describe the soil conditions of each site (pairs of plots). For further analyses we considered pH, total N and total C obtained using methods described in Kurek et al. (2014a).

To determine the total dissimilarity between bryophyte assemblages, respective components of turnover (species replacement between sites) and nestedness (species loss from site to site), we performed a multiple-site dissimilarity analysis (Baselga and Orme, 2012) using the “beta.multi” function in the “betapart” package (Baselga et al., 2013). To quantify and compare the diversity of bryophyte species between burrows and reference plots, we calculated the Shannon H' index of diversity according to the formula $H' = -\sum_{i=1}^S p_i \ln p_i$, where S is species richness and p_i is the ratio of individuals belonging to the i th species. Indicator species analyses (Dufréne and Legendre, 1997) using the “indval” function in the “labdsv” package (Roberts, 2012) were performed to reveal the preference of a species for the plot type (disturbed – burrow; undisturbed – reference plot).

Statistical analyses and index calculations employed Statistica 9 for ANOVA tests, CANOCO 5 for DCA analysis, and R v2.13.1 for the multiple-site dissimilarity analysis, indicator species analyses and chi square test (R Development Core Team, 2009). Prior to the statistical tests, the data for ANOVA were transformed with logarithmic or exponential functions to obtain a normal or at least

symmetric distribution. All the data were analyzed in two ways: jointly, and divided into two groups depending on habitat fertility, classified using the C/N ratio, with all the plots having C/N > 15 classified as stands with lower fertility. One-way ANOVA was applied to determine the significance of differences between habitats. Split-plot ANOVA was used to examine the effect of habitat fertility (lower-fertility habitats with C/N > 15 vs. higher-fertility habitats with C/N < 15) and of disturbance level (burrow vs. reference) on mean number of species and Shannon H' . The chi square test was used to check the significance of any differences between plot types as to the ratios of the number of bryophyte species representing different functional groups (tables 2 × 3 and 2 × 5).

An indirect gradient analysis performed with detrended correspondence analysis (DCA with default settings, without transformation and downweighting of rare species) was used to disclose relationships between plot types and to identify the main gradients of bryophyte occurrence across plot types. Bryophyte species occurring in only one plot (accidental records) were excluded from further analysis in order to eliminate the potential influence of sporadically occurring species.

3. Results

A total of 55 bryophyte species were recorded in the survey. Of these, 26 of the 49 (53.1%) species recorded in burrow plots occurred exclusively on burrows; 6 of the 29 (20.7%) species recorded in reference plots occurred exclusively there. Twenty-three species occurred in both burrow and reference plots (Table 1). Ten of the 26 (38.5%) species occurring exclusively on burrows were identified as epiphytic bryophytes on the bark of fleshy-fruited shrubs (mostly *Sambucus nigra*) growing on badger burrows (*Amblystegium serpens*, *Bryum caespiticium*, *Leskea polycarpa*, *Leskella nervosa*, *Mnium spinosum*, *Orthotrichum affine*, *Orthotrichum diaphanum*, *Pteryginandrum filiforme*, *Pylaisia polyantha*, *Radula complanata*).

Mean number of bryophyte species and Shannon H' were significantly higher for burrow plots (hypothesis 1) than for reference plots ($F = 23.93$, $P = 0.0001$; $F = 21.46$, $P = 0.0002$; Fig. 1). The C/N ratios, expressing habitat fertility, confirmed our division of the plots into two habitat fertility groups: lower fertility (C/N > 15) and higher fertility (C/N < 15). There were no significant differences between environmental characteristics depending on habitat fertility (hypothesis 2, Table 2). Nor was the habitat fertility × disturbance effect significant for number of species ($F = 0.65$, $P = 0.4304$) or for Shannon H' ($F = 2.54$, $P = 0.1285$). Total dissimilarity was similar for setts and reference plots (β_{SOR} for setts = 0.90, β_{SOR} for reference = 0.92). Turnover (β_{SIM} for setts = 0.85, β_{SIM} for reference = 0.83) and nestedness (β_{SNE} for setts = 0.05, β_{SNE} for reference = 0.09) for the bryophyte assemblages of burrows and reference plots were also similar.

There were no differences between burrows and reference plots in the abundance of bryophytes representing specific substrates, acidity requirements, or specific life histories (hypothesis 3). No group of species growing on specific substrates was significantly overrepresented ($\chi^2 = 4.299$, $df = 4$, $P = 0.367$, Fig. 2), but it is worth noting that epiphytic bryophytes were represented by 14 species in burrow plots and by only 3 species in reference plots. For acidity, higher (but not significantly higher) shares of taxa ranging from m acidophytes (moderately acidic) to basiphytes were recorded on burrows ($\chi^2 = 0.864$, $df = 4$, $P = 0.930$). Some species requiring higher-pH substrate were significantly more frequent on burrows: *Plagiomnium rostratum* and *Atrichum undulatum* (Table 1). The shares of annuals, short-lived shuttle species and perennials on

Table 1

Bryophyte species present in burrow plots and reference plots, their occurrence across the 40 plots, indicator values, and the significance level P associated with the maximum indicator value. Significant indicator values are bolded. Asterisks denote epiphytic species occurring exclusively on trunks of endozoochorous plant species growing on burrows.

Species	Occurrence	Indicator values		P
		Burrow (N = 20)	Reference (N = 20)	
<i>Plagiomnium rostratum</i>	16	53.5	2.2	0.005
<i>Atrichum undulatum</i>	12	47.1	0.7	0.006
<i>Dicranella heteromalla</i>	9	45.0	0	0.001
<i>Hypnum cupressiforme</i>	13	39.3	3.2	0.035
<i>Lophocolea heterophylla</i>	10	32.0	2.0	0.047
<i>Pseudoscleropodium purum</i>	6	30.0	0	0.027
<i>Pohlia nutans</i>	15	27.0	12.0	n.s.
<i>Polytrichastrum formosum</i>	16	27.0	14.0	n.s.
<i>Brachythecium salebrosum</i>	14	26.4	9.4	n.s.
<i>Plagiothecium curvifolium</i>	9	25.3	2.4	n.s.
<i>Hypnum cupressiforme</i> var. <i>filiforme</i>	5	25.0	0	0.049
<i>Amblystegium serpens</i> *	5	25.0	0	0.045
<i>Brachythecium rutabulum</i>	7	20.7	1.7	n.s.
<i>Plagiothecium denticulatum</i>	6	16.0	2.0	n.s.
<i>Aulacomnium androgynum</i>	8	15.6	5.6	n.s.
<i>Plagiomnium affine</i>	6	15.4	2.3	n.s.
<i>Lophocolea bidentata</i>	3	15.0	0	n.s.
<i>Orthotrichum diaphanum</i> *	3	15.0	0	n.s.
<i>Pohlia elongata</i>	3	15.0	0	n.s.
<i>Pseudotaxiphyllum elegans</i>	3	15.0	0	n.s.
<i>Eurhynchium striatum</i>	15	13.5	21.0	n.s.
<i>Ceratodon purpureus</i>	6	13.3	3.3	n.s.
<i>Brachytheciastrum velutinum</i>	2	10.0	0	n.s.
<i>Herzogiella seligeri</i>	2	10.0	0	n.s.
<i>Leskea polycarpa</i> *	2	10.0	0	n.s.
<i>Orthotrichum affine</i> *	2	10.0	0	n.s.
<i>Pteryginandrum filiforme</i> *	2	10.0	0	n.s.
<i>Pleurozium schreberi</i>	22	9.1	51.9	0.041
<i>Dicranum scoparium</i>	8	0.6	28.1	n.s.
<i>Dicranum polysetum</i>	5	0.9	13.7	n.s.
<i>Orthodicranum montanum</i>	2	0	10.0	n.s.
<i>Ptilium crista-castrensis</i>	2	0	10.0	n.s.
<i>Sanionia uncinata</i>	2	0	10.0	n.s.
<i>Hylacomium splendens</i>	6	5.1	9.9	n.s.
<i>Dicranodontium denundatum</i>	1	0	5.0	n.s.
<i>Ditrichum heteromallum</i>	1	0	5.0	n.s.
<i>Funaria hygrometrica</i>	1	0	5.0	n.s.
<i>Bryum pseudotriquetrum</i>	2	2.5	2.5	n.s.
<i>Buxbaumia aphylla</i>	2	2.5	2.5	n.s.
<i>Plagiothecium laetum</i>	2	2.5	2.5	n.s.
<i>Tetraphis pellucida</i>	3	6.7	1.7	n.s.
<i>Plagiomnium undulatum</i>	3	7.5	1.3	n.s.
<i>Bryum caespiticium</i> *	1	5.0	0	n.s.
<i>Calypogeia azurea</i>	1	5.0	0	n.s.
<i>Calypogeia suecica</i>	1	5.0	0	n.s.
<i>Cephaloziella varians</i>	1	5.0	0	n.s.
<i>Hypnum andoi</i>	1	5.0	0	n.s.
<i>Isoetecium alopecuroides</i>	1	5.0	0	n.s.
<i>Leskella nervosa</i> *	1	5.0	0	n.s.
<i>Mnium spinosum</i> *	1	5.0	0	n.s.
<i>Plagiomnium ellipticum</i>	1	5.0	0	n.s.
<i>Plagiothecium nemorale</i>	1	5.0	0	n.s.
<i>Pylaisia polyantha</i> *	1	5.0	0	n.s.
<i>Radula complanata</i> *	1	5.0	0	n.s.
<i>Rhytidiadelphus subpinnatus</i>	1	5.0	0	n.s.

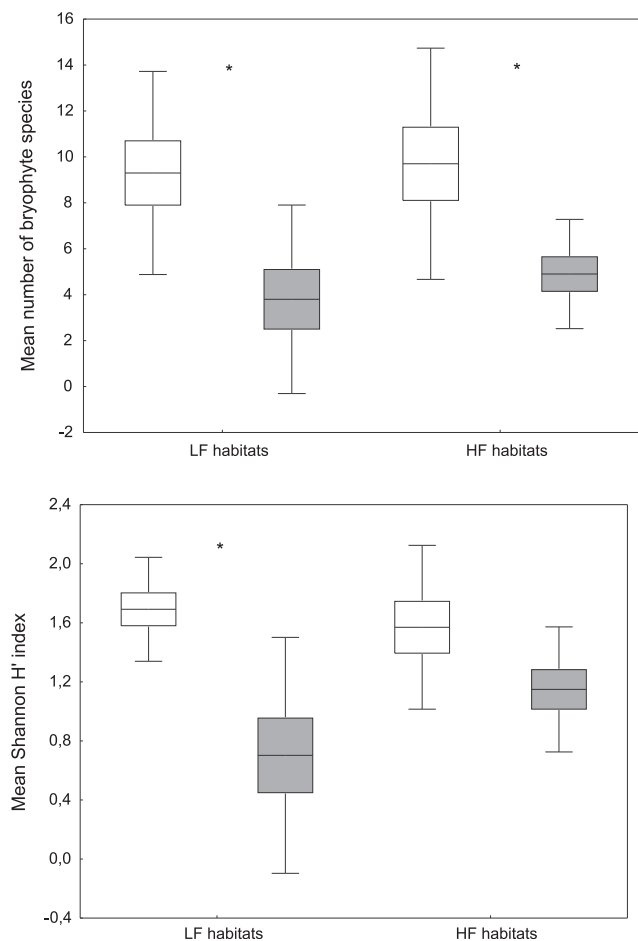


Fig. 1. Parameters of bryophyte communities in habitats with lower fertility (LF habitats, C/N > 15) and higher fertility (HF habitats, C/N < 15), determined for burrows and reference plots. Solid line – means; box – standard error; whiskers – standard deviation. White – burrows; gray – reference plots. * $P < 0.05$.

burrows and in reference plots were similar ($\chi^2 = 1.163$, $df = 2$, $P = 0.559$, Fig. 2).

In the indicator species analysis, 8 species (14.5%, $N = 55$) were significantly associated with burrows and occurred mostly or even exclusively on them (hypothesis 3, Table 1). Among the taxa of the genus *Plagiomnium*, *P. rostratum* in particular dominated the majority of the burrow plots, growing on the bare ground of spoil heaps. The epiphytes mentioned above also occurred mostly on burrows, but only the abundance of *A. serpens* was high enough to reach statistical significance. Only one taxon, *Pleurozium schreberi*, a typical species of the ground layer in pine forests, was associated chiefly with undisturbed reference plots. The DCA analysis also indicated some qualitative differences between the burrows and reference plots. In general the analysis yielded a mixed group of plots (Fig. 3), but there was some support for distinguishing two groups of bryophyte species: bryophyte communities associated

with reference plots, and bryophyte communities associated with badger setts (a less separated group, partly intermingled with other plots). One main gradient seems to emerge: a topsoil disturbance regime gradient running from species characteristic of an undisturbed ground layer in forests (*Dicranum polysetum*, *Dicranum scoparium*, *P. schreberi*, *Hylocomium splendens*, *Ptilium crista-castrensis*; at left in Fig. 3), to species requiring bare substrate, that is, bare ground after topsoil disturbance *P. rostratum* or the bark of fleshy-fruited shrubs (*L. polycarpa*, *O. diaphanum*, *P. filiforme*; upper right in Fig. 3).

4. Discussion

As shown elsewhere in the literature, digging activity by badgers among their burrows plays a role by producing small-scale disturbances which can substantially influence the local floristic composition quantitatively and qualitatively (Kurek et al., 2014a; Obidziński and Głogowski, 2005; Obidziński and Kiełtyk, 2006). Our results are in line with those findings. The burrows in our study had higher species richness than the reference plots, and hosted 49 species – 28.8% of the 170 bryophyte species occurring within the study area (Kampinos National Park – unpublished database). Quantitative partitioning of burrows and reference plots occurred regardless of habitat fertility (C/N ratio). These findings confirm the hypothesis that badger digging quantitatively influences the compositional heterogeneity of bryophytes.

Bryophytes are divided into functional types based on a system of coevolved adaptive traits that is used to characterize organisms, communities, and the particular ecological niches they occupy (Kürschner, 2004). These adaptive traits are based mainly on complex life forms (Mägdefrau, 1982), life strategies (During, 1979; Glime, 2013) and ecomorphological adaptations (Kürschner, 2004) which arise convergently but independently in quite unrelated taxa when they evolve under similar habitat conditions (Kürschner, 2004). With the exception of some epiphytes, we found no connection between bryophyte functional groups and specific habitat types associated with the bare ground of badger burrows. One possible explanation is that bryophytes have a wide ecological amplitude; the same species may occur in different habitats and on different substrates (see review in Dierssen, 2001). Only epiphytes growing on the stems of fleshy-fruited shrubs and trees (e.g. *S. nigra*, *Pyrus* sp., *Cornus sanguinea*) seem to be a characteristic group of bryophytes associated mostly with burrows. Fleshy-fruited shrubs and trees are more frequent there as a result of badger activity (Kurek et al., 2014a). Epiphytic bryophytes colonize the bark of fleshy-fruited shrubs and trees. That is a secondary effect of the shaping of diversity by burrowing badgers.

In the case of herbaceous plants, disturbance can promote species with a shorter life history, especially annuals (Kinlaw, 1999; Obidziński and Głogowski, 2005; Symonides, 1988), a group highly resistant to any ground disturbance. This is not an important trait in bryophytes. We did not record a higher share of annuals or at least short-lived shuttle species and colonists on the burrows. We offer two reasons for this: annuals and short-lived shuttle

Table 2
Results from one-way ANOVA of differences in habitat parameters between habitats with lower fertility (LF habitats, C/N > 15) and higher fertility (HF habitats, C/N < 15). BA – stand basal area. Asterisk denotes statistical significance.

	LF habitats				HF habitats				F	P
	Mean	SD	Min	Max	Mean	SD	Min	Max		
BA of Scots pine (%)	76.7	17.7	48.3	100.0	66.0	42.6	0.0	100.0	0.06	0.8126
N (%)	0.086	0.031	0.043	0.135	0.104	0.093	0.035	0.359	0.00	0.9760
C (%)	1.660	0.525	0.801	2.351	1.287	0.903	0.403	3.623	2.49	0.1318
C/N*	19.6	2.7	15.4	24.0	13.1	1.5	10.1	14.8	48.02	<0.0001
pH	5.03	0.30	4.59	5.46	4.96	0.23	4.67	5.37	0.34	0.5694

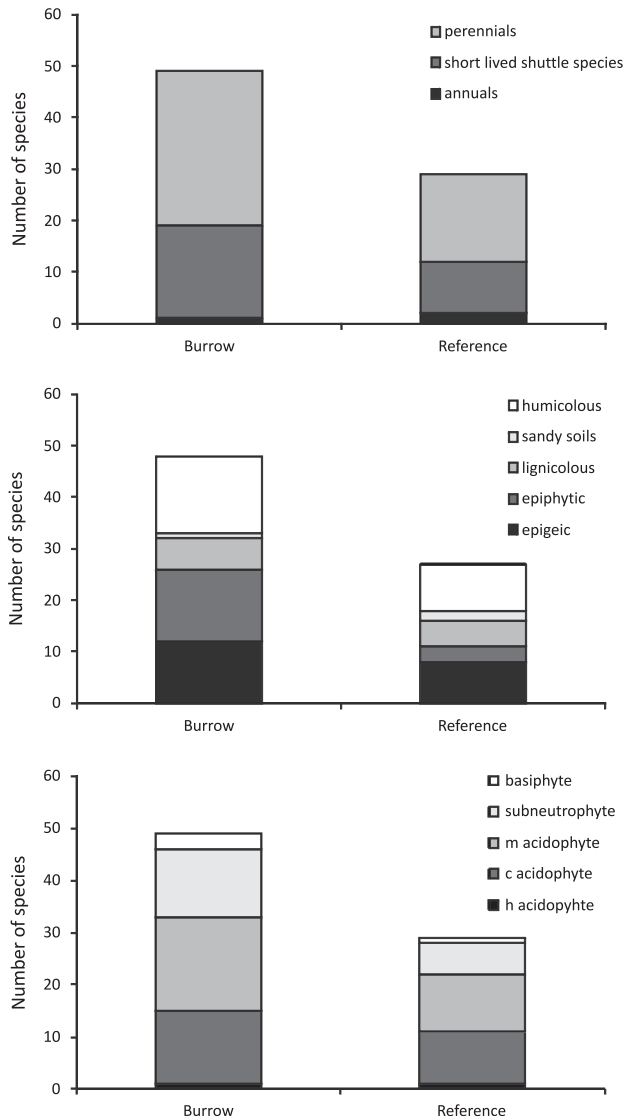


Fig. 2. Number of bryophyte species occurring in burrow plots and reference plots, by life history (a), substrate type (b) and acidity requirement (c).

species prefer habitats radically different from the forest floor and thus have low representation on the forest floor (During, 1979, 1992; Glime, 2013); and in annuals and short-lived bryophytes (long spore life span) adaptive traits may not be important enough to allow them to colonize and establish on disturbed ground in forest conditions (During, 1979, 1992; Glime, 2013; Kürschner, 2004).

A strong correlation between life form, life cycle strategy, community structure and site environmental conditions has been demonstrated repeatedly (e.g. Bates, 1998; Gimingham and Birse, 1957; Glime, 2013; Kürschner, 2004; Lonnel et al., 2014; Mägdefrau, 1982; Økland et al., 2003). Some authors report that especially competitive perennials (e.g. *A. undulatum* or *Brachythecium rutabulum*) and long-lived shuttle species (e.g. *Plagiomnium* spp.) tend to dominate the forest floor (Darell and Cronberg, 2011; Glime, 2013). Bare soil is colonized by competitive perennials and long-lived shuttle species (Darell and Cronberg, 2011), as evidenced in our research, and not necessarily by annuals and short-lived bryophytes. Species adopting the shuttle strategy have a competitive advantage, as they occupy recurring sites by producing large spores in diaspore banks, which provide a long-term safeguard against unfavorable conditions (Kürschner, 2004). Diaspore

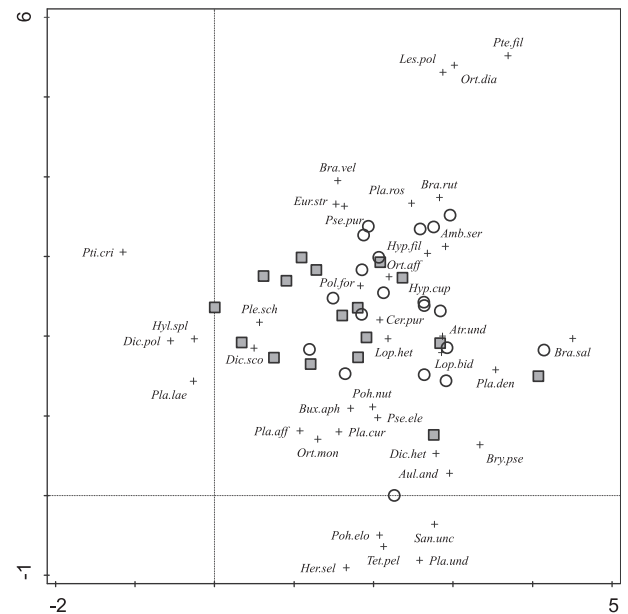


Fig. 3. Detrended correspondence analysis (DCA) scatterplot showing the relationship between plot type and bryophyte composition for 40 study plots on the first two axes. Legend: circles – burrows; squares – reference plots. Abbreviations of species: Amb.ser – *Amblystegium serpens*, Atr.und – *Atrichum undulatum*, Aul.and – *Aulaconium androgynum*, Bra.vel – *Brachytheciastrum velutinum*, Bra.sal – *Brachythecium salebrosum*, Bra.rut – *Brachythecium rutabulum*, Bry.pse – *Bryum pseudotriquetrum*, Bux.aph – *Buxbaumia aphylla*, Cer.pur – *Ceratodon purpureus*, Dic.het – *Dicranella heteromalla*, Dic.pol – *Dicranum polysetum*, Dic.sco – *Dicranum scoparium*, Eur.str – *Eurhynchium striatum*, Her.sel – *Herzogiella seligeri*, Hyl.spl – *Hylocomium splendens*, Hyp.cup – *Hypnum cupressiforme*, Hyp.fil – *Hypnum cupressiforme* var. *filiforme*, Les.pol – *Leskea polycarpa*, Lop.bid – *Lophocolea bidentata*, Lop.het – *Lophocolea heterophylla*, Ort.mon – *Orthodictyon montanum*, Ort.aff – *Orthotrichum affine*, Ort.dia – *Orthotrichum diaphanum*, Pla.aff – *Plagiomnium affine*, Pla.ros – *Plagiomnium rostratum*, Pla.und – *Plagiomnium undulatum*, Pla.cur – *Plagiothecium curvifolium*, Pla.den – *Plagiothecium denticulatum*, Pla.lae – *Plagiothecium laetum*, Ple.sch – *Pleurozium schreberi*, Poh.nut – *Pohlia nutans*, Poh.elo – *Pohlia elongata*, Pol.for – *Polytrichastrum formosum*, Pse.pur – *Pseudoscleropodium purum*, Pse.ela – *Pseudotaxiphyllum elegans*, Pte.fil – *Pteryginandrum filiforme*, Pti.cri – *Ptilium crista-castriensis*, San.unc – *Sanionia uncinata*, Tet.pel – *Tetraphis pellucida*.

banks bridge the gaps between favorable and unfavorable periods and lower the risk of local extinction (During, 1992; Kürschner, 2004; Longton, 1997); this is the reason for their resistance to cyclical soil disturbance. When topsoil disturbances are scarce or not extensive, perennial stayers may also benefit and establish on bare spoil heaps.

Sensitive species and species not resistant to competition from other bryophytes and herbaceous plants may prefer to colonize burrows because they offer large areas of bare ground with lower competitive pressure. Bryophytes are much more difficult to classify in a specific group, as we show in the results. Apart from epiphytes, some bryophytes presented in Tables 1 and 2 require bare ground, especially *P. rostratum*, *Dicranella heteromalla*, *A. undulatum* and *Polytrichastrum formosum*. This seems to be a common feature which explains their higher frequency on burrows. Bare ground is a characteristic of disturbed areas, and disturbances of the soil surface at badger burrows are an important factor in the occurrence of bryophytes there. Release from competition explains the higher richness of herbaceous plants in disturbed areas, including burrows (Kinlaw, 1999; Kyle et al., 2007; Wilson and Tilman, 1993). Bryophytes requiring higher-pH substrate find more suitable conditions for colonization on burrows (i.e. *P. rostratum*, *A. undulatum*). Spoil heaps produced by digging animals have higher pH (Canals and Sebastia, 2000; Eldridge and Myers, 2001; Kerley et al., 2004; Kurek et al., 2014a) than adjacent areas where the topsoil is

acidulated by products of tree litter decomposition, microbial respiration and plant root exudates (Jobbágy and Jackson, 2003).

Our findings concerning the characteristic features of the investigated bryophyte assemblages also confirm qualitative changes occurring on badger burrows, but in some cases it was difficult to directly estimate the qualitative impact of badger burrowing on bryophyte assemblages, because we obtained similar values for total dissimilarity, turnover and nestedness of bryophytes, and close relationships (DCA) between the setts and reference plots.

As the topsoil disturbance caused by badger digging creates niches suitable for bryophyte settlement, the spatial extent of badger activity should be determined. Each social group of badgers possesses and utilizes several shelters (setts) in their territory (Feore and Montgomery, 1999), which vary in their number of entrances and chambers and, as a result, in the size of the disturbed area. Badger family life, rearing of cubs and winter sleep are focused in the main setts, which are well developed, have many entrances and a large amount of excavated soil, and are inherited by successive generations (Goszczyński and Wójtowicz, 2001; Kowalczyk et al., 2004). The density of the badger's main setts varies considerably along its European range, from 0.4 setts/10 km² in the Białowieża Primeval Forest (Poland) to over 45 setts/10 km² in some regions of England and Ireland (Kowalczyk et al., 2000). Except for the Białowieża Primeval Forest, the density of main setts in Poland ranges from 1.0 to 1.1 setts/10 km² (Goszczyński, 1999; Goszczyński and Skoczyska, 1996). The effect of badger-caused disturbance of topsoil is spatially wider, as badgers also have more or less developed burrows other than their main setts (subsidiary setts, annexes) in their territories (Feore and Montgomery, 1999). The size of a disturbed area ranges from tens to hundreds of square meters (Coombes and Viles, 2015; Obidziński et al., 2013) in well-developed and long-term setts. The sandy heaps and trodden ground of large, elaborate badger setts vary in size; the volume of removed soil placed on the surface may even reach 27.8 m³ (Coombes and Viles, 2015; Neal and Roper, 1991).

Disturbances in Central European temperate forests are minimized mainly by forest management practices (e.g. removal of dying trees which would disturb the soil when they uproot). This also applies to forests in protected areas: on average only 17.2% of Poland's lowland national parks are under strict protection and not subjected to forest management practices (Statistical Yearbook of the Republic of Poland, 2010). Badger burrowing in managed forests (and also in national parks) may be an important factor boosting the diversity of bryophyte species, especially where natural topsoil disturbances are limited by silviculture methods. In forest ecosystems, badger activity creates a patch distribution of microhabitats for vascular plants and, as we showed here, for bryophytes. In Central Europe, badgers prefer to settle in safe, secluded places (Kurek, 2011). Badgers should not be classified as game animals in protected areas (national parks, reserves and other protected areas), nor outside of those areas in managed forests. Research on mortality in a local population in the Western Carpathians showed that hunters are an important factor in badger mortality (Mysłajek et al., 2012). Excessive hunting can diminish the effect of badger activity as a seed disperser and a habitat engineer in both managed and protected forest ecosystems. Eliminating culling pressure on badger populations may have a positive impact on plant diversity; only well-developed, long-term setts can provide enough suitable microhabitats for herbaceous plants, shrubs, trees (Kurek et al., 2014a, 2014b) – and bryophytes – to colonize.

The majority of quantitative and qualitative traits of bryophyte assemblages on badger setts differed from the surrounding areas, confirming the importance of badgers to the compositional hetero-

geneity of bryophytes in forested areas. The spatial scale and the disturbance range of badger digging make badger setts a significant factor shaping the species diversity of bryophytes (and vascular plants) in Central European managed and protected forests, and probably throughout the range of the species.

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